

```
!pip install boto3 sagemaker torch torchvision matplotlib
```

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Requirement already satisfied: boto3 in /usr/local/lib/python3.12/dist-packages (1.40.69)
Requirement already satisfied: sagemaker in /usr/local/lib/python3.12/dist-packages (2.254.1)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.8.0+cu126)
Requirement already satisfied: torchvision in /usr/local/lib/python3.12/dist-packages (0.23.0+cu126)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: botocore<1.41.0,>=1.40.69 in /usr/local/lib/python3.12/dist-packages (from boto3) (1.40.69)
Requirement already satisfied: jmespath<2.0.0,>=0.7.1 in /usr/local/lib/python3.12/dist-packages (from boto3) (1.0.1)
Requirement already satisfied: s3transfer<0.15.0,>=0.14.0 in /usr/local/lib/python3.12/dist-packages (from boto3) (0.14.0)
Requirement already satisfied: attrs<26,>=24 in /usr/local/lib/python3.12/dist-packages (from sagemaker) (25.4.0)
Requirement already satisfied: cloudpickle>=2.2.1 in /usr/local/lib/python3.12/dist-packages (from sagemaker) (3.1.2)
Requirement already satisfied: docker in /usr/local/lib/python3.12/dist-packages (from sagemaker) (7.1.0)
Requirement already satisfied: fastapi in /usr/local/lib/python3.12/dist-packages (from sagemaker) (0.121.0)
Requirement already satisfied: google-pasta in /usr/local/lib/python3.12/dist-packages (from sagemaker) (0.2.0)
Requirement already satisfied: graphene<4,>=3 in /usr/local/lib/python3.12/dist-packages (from sagemaker) (3.4.3)
Requirement already satisfied: importlib-metadata<7.0,>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from sagemaker) (0.121.0)
Requirement already satisfied: jsonschema in /usr/local/lib/python3.12/dist-packages (from sagemaker) (4.25.1)
Requirement already satisfied: numpy<3.0,>=1.26.4 in /usr/local/lib/python3.12/dist-packages (from sagemaker) (2.0.2)
Requirement already satisfied: omegaconf<3,>=2.2 in /usr/local/lib/python3.12/dist-packages (from sagemaker) (2.3.0)
Requirement already satisfied: packaging<25,>=23.0 in /usr/local/lib/python3.12/dist-packages (from sagemaker) (24.2)
Requirement already satisfied: pandas>=2.3.0 in /usr/local/lib/python3.12/dist-packages (from sagemaker) (2.3.3)
Requirement already satisfied: pathos in /usr/local/lib/python3.12/dist-packages (from sagemaker) (0.3.4)
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Requirement already satisfied: protobuf<6.32,>=3.12 in /usr/local/lib/python3.12/dist-packages (from sagemaker) (5.29.5)
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Requirement already satisfied: pyyaml>=6.0.1 in /usr/local/lib/python3.12/dist-packages (from sagemaker) (6.0.3)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from sagemaker) (2.32.4)
Requirement already satisfied: sagemaker-core<2.0.0,>=1.0.17 in /usr/local/lib/python3.12/dist-packages (from sagemaker) (1.0.17)
Requirement already satisfied: schema in /usr/local/lib/python3.12/dist-packages (from sagemaker) (0.7.8)
Requirement already satisfied: smdebug-rulesconfig==1.0.1 in /usr/local/lib/python3.12/dist-packages (from sagemaker) (1.0.1)
Requirement already satisfied: tblib<4,>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from sagemaker) (3.2.1)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from sagemaker) (4.67.1)
Requirement already satisfied: urllib3<3.0.0,>=1.26.8 in /usr/local/lib/python3.12/dist-packages (from sagemaker) (2.5.0)
Requirement already satisfied: uvicorn in /usr/local/lib/python3.12/dist-packages (from sagemaker) (0.38.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch) (3.20.0)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.13.3)
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Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.6.77 in /usr/local/lib/python3.12/dist-packages (from torch) (12.6.77)
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.6.77 in /usr/local/lib/python3.12/dist-packages (from torch) (12.6.77)
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Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in /usr/local/lib/python3.12/dist-packages (from torch) (9.10.2.21)
Requirement already satisfied: nvidia-cublas-cu12==12.6.4.1 in /usr/local/lib/python3.12/dist-packages (from torch) (12.6.4.1)
Requirement already satisfied: nvidia-cufft-cu12==11.3.0.4 in /usr/local/lib/python3.12/dist-packages (from torch) (11.3.0.4)
Requirement already satisfied: nvidia-curand-cu12==10.3.7.77 in /usr/local/lib/python3.12/dist-packages (from torch) (10.3.7.77)
Requirement already satisfied: nvidia-cusolver-cu12==11.7.1.2 in /usr/local/lib/python3.12/dist-packages (from torch) (11.7.1.2)
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Requirement already satisfied: nvidia-nccl-cu12==2.27.3 in /usr/local/lib/python3.12/dist-packages (from torch) (2.27.3)
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Requirement already satisfied: pillow!=8.3.*,>=5.3.0 in /usr/local/lib/python3.12/dist-packages (from torchvision) (11.3.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
```

```
import time
import numpy as np
import torch
import matplotlib.pyplot as plt
import boto3
import sagemaker
from sagemaker.predictor import Predictor
```

```
sagemaker.config INFO - Not applying SDK defaults from location: /etc/xdg/sagemaker/config.yaml
sagemaker.config INFO - Not applying SDK defaults from location: /root/.config/sagemaker/config.yaml
```

```
import os

os.environ["AWS_ACCESS_KEY_ID"] = "AKIAXWNGFDSY72KTJI6G"
os.environ["AWS_SECRET_ACCESS_KEY"] = "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"
os.environ["AWS_DEFAULT_REGION"] = "us-east-1" # change if needed

session = sagemaker.Session()
role = "arn:aws:sagemaker:us-east-1:529166310577:notebook-instance/lab8"

print("✅ AWS setup complete!")
```

✅ AWS setup complete!

```
import torch.nn as nn

# Example dummy model for demo
class SimpleModel(nn.Module):
    def __init__(self):
        super(SimpleModel, self).__init__()
        self.fc = nn.Linear(10, 1)

    def forward(self, x):
        return self.fc(x)

# Load model
model = SimpleModel()
model.eval()

# Example input
x_test = torch.randn(1, 10)
```

```
latencies = []
for _ in range(20):
    start = time.time()
    with torch.no_grad():
        _ = model(x_test)
    end = time.time()
    latencies.append((end - start) * 1000)

local_avg_latency = np.mean(latencies)
print(f"🟡 Local Inference Latency: {local_avg_latency:.2f} ms")
```

🟡 Local Inference Latency: 2.77 ms

```
endpoint_name = "xgb-fixed-1762756784-endpoint"

predictor = Predictor(endpoint_name=endpoint_name, sagemaker_session=session)
print(f"✅ Connected to SageMaker endpoint: {endpoint_name}")
```

✅ Connected to SageMaker endpoint: xgb-fixed-1762756784-endpoint

```
import boto3

sm_client = boto3.client('sagemaker')
response = sm_client.list_endpoints(SortBy='CreationTime', SortOrder='Descending')

for ep in response['Endpoints']:
    print(f"Name: {ep['EndpointName']} | Status: {ep['EndpointStatus']}")
```

Name: xgb-fixed-1762756784-endpoint | Status: InService
Name: xgboost-loadtest-1762754915-endpoint | Status: Failed

```
print("Payload being sent:")
print(payload_csv)
```

Payload being sent:
1.3863403797149658,-0.6994120478630066,0.3085666000843048,-1.0522058010101318,1.3556939363479614,-1.5536372661590576,-0.0536

```
payload_csv = ",".join(map(str, np.ravel(x_test.tolist()))) + "\n"
```

```
import numpy as np

# Example: model expects 100 features
expected_features = 100
x_test = np.random.randn(1, expected_features) # single row, 100 features
payload_csv = ",".join(map(str, np.ravel(x_test.tolist()))) + "\n"
```

```
import boto3
import re

# Connect to CloudWatch
logs = boto3.client('logs', region_name='us-east-1')

# Your log group name
log_group = '/aws/sagemaker/Endpoints/xgb-fixed-1762756784-endpoint'

# Get latest log streams
streams = logs.describe_log_streams(
    logGroupName=log_group,
    orderBy='LastEventTime',
    descending=True,
    limit=3
)['logStreams']

# Read last 100 log events from the newest stream
for s in streams:
    events = logs.get_log_events(
        logGroupName=log_group,
        logStreamName=s['logStreamName'],
        limit=100
    )['events']

    print(f" ♦ Checking stream: {s['logStreamName']}\n")
    for e in events:
        msg = e['message']
        if any(x in msg for x in ['ValueError', 'expected', 'Traceback', 'mismatch']):
            print(msg)
```

♦ Checking stream: AllTraffic/i-07e7815677cad873b

```
import boto3
logs = boto3.client('logs', region_name='us-east-1')

log_group = '/aws/sagemaker/Endpoints/xgb-fixed-1762756784-endpoint'
stream_name = 'AllTraffic/i-07e7815677cad873b'

# Fetch more logs
events = logs.get_log_events(
    logGroupName=log_group,
    logStreamName=stream_name,
    limit=2000,
    startFromHead=True
)['events']

found = False
for e in events:
    msg = e['message']
    if any(x in msg for x in ['ValueError', 'expected', 'Traceback', 'mismatch']):
        print(" ⚠️ ", msg)
        found = True

if not found:
    print("❌ No ValueError or mismatch found yet. Try re-running your Colab inference cell again, then re-run this log-ch
```

⚠️ Traceback (most recent call last):
File "/miniconda3/lib/python3.8/site-packages/sagemaker_xgboost_container/algorithm_mode/serve.py", line 226, in invocat
preds = ScoringService.predict(data=dtest, content_type=content_type, model_format=format)
File "/miniconda3/lib/python3.8/site-packages/sagemaker_xgboost_container/algorithm_mode/serve.py", line 77, in predict
return serve_utils.predict(cls.booster, model_format, data, content_type, cls.objective)
File "/miniconda3/lib/python3.8/site-packages/sagemaker_xgboost_container/algorithm_mode/serve_utils.py", line 235, in p
return model.predict(dtest, ntree_limit=getattr(model, "best_ntree_limit", 0), validate_features=False)
File "/miniconda3/lib/python3.8/site-packages/xgboost/core.py", line 1913, in predict
_check_call(
File "/miniconda3/lib/python3.8/site-packages/xgboost/core.py", line 218, in _check_call
raise XGBoostError(py_str(_LIB.XGBGetLastError()))
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Traceback (most recent call last):
File "/miniconda3/lib/python3.8/site-packages/sagemaker_xgboost_container/algorithm_mode/serve.py", line 214, in invocat
dtest, content_type = serve_utils.parse_content_data(payload, flask.request.content_type)
File "/miniconda3/lib/python3.8/site-packages/sagemaker_xgboost_container/algorithm_mode/serve_utils.py", line 118, in p
content_type = get_content_type(input_content_type)
File "/miniconda3/lib/python3.8/site-packages/sagemaker_xgboost_container/data_utils.py", line 120, in get_content_type
    raise exc.UserError(_get_invalid_content_type_error_msg(content_type_cfg_val))
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File "/miniconda3/lib/python3.8/site-packages/sagemaker_xgboost_container/algorithm_mode/serve_utils.py", line 235, in p

```

```

import boto3

logs = boto3.client('logs', region_name='us-east-1')
log_group = '/aws/sagemaker/Endpoints/xgb-fixed-1762756784-endpoint'
stream_name = 'AllTraffic/i-07e7815677cad873b'

events = logs.get_log_events(
    logGroupName=log_group,
    logStreamName=stream_name,
    limit=3000,
    startFromHead=True
)['events']

# capture traceback + next few lines
for i, e in enumerate(events):
    msg = e['message']
    if 'XGBoostError' in msg or 'ValueError' in msg:
        print("▲", msg)
        # print 3 lines after error for context
        for j in range(1, 4):
            if i + j < len(events):
                print("  ", events[i + j]['message'])
        print("-" * 80)

```

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l xgboost.core.XGBoostError: [06:44:12] ../src/learner.cc:1257: Check failed: learner_model_param_.num_feature >= p_fma
l Stack trace:
[bt] (0) /miniconda3/lib/python3.8/site-packages/xgboost/lib/libxgboost.so(+0x19cab9) [0x7fd82e1e9ab9]
[bt] (1) /miniconda3/lib/python3.8/site-packages/xgboost/lib/libxgboost.so(+0x1aa8ad) [0x7fd82e1f78ad]
[bt] (2) /miniconda3/lib/python3.8/site-packages/xgboost/lib/libxgboost.so(+0x1aa93d) [0x7fd82e1f793d]
[bt] (3) /miniconda3/lib/python3.8/site-packages/xgboost/lib/libxgboost.so(+0x1aac2d) [0x7fd82e1f7c2d]
[bt] (4) /miniconda3/lib/python3.8/site-packages/xgboost/lib/libxgboost.so(XGBoosterPredictFromDMatrix+0x346) [0x7fd82e0

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```
[bt] (5) /miniconda3/lib/python3.8/lib-dynload/../../libffi.so.8(+0xa052) [0x7fd8647ff052]
[bt] (6) /miniconda3/lib/python3.8/lib-dynload/../../libffi.so.8(+0x8925) [0x7fd8647fd925]
[bt] (7) /miniconda3/lib/python3.8/lib-dynload/../../libffi.so.8(ffi_call+0xde) [0x7fd8647fe06e]
[bt] (8) /miniconda3/lib/python3.8/lib-dynload/_ctypes.cpython-38-x86_64-linux-gnu.so(_ctypes_callproc+0x9d2) [0x7fd8648
  169.254.178.2 - - [10/Nov/2025:06:44:12 +0000] "POST /invocations HTTP/1.1" 400 1200 "-" "AHC/2.0"
```

```

xgboost.core.XGBoostError: [06:44:12] ../src/learner.cc:1257: Check failed: learner_model_param_.num_feature >= p_fmat-
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  169.254.178.2 - - [10/Nov/2025:06:44:12 +0000] "POST /invocations HTTP/1.1" 400 1200 "-" "AHC/2.0"
  169.254.178.2 - - [10/Nov/2025:06:44:14 +0000] "GET /ping HTTP/1.1" 200 0 "-" "AHC/2.0"
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```

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  Stack trace:
[bt] (0) /miniconda3/lib/python3.8/site-packages/xgboost/lib/libxgboost.so(+0x19cab9) [0x7fd82e1e9ab9]
[bt] (1) /miniconda3/lib/python3.8/site-packages/xgboost/lib/libxgboost.so(+0x1aa8ad) [0x7fd82e1f78ad]
[bt] (2) /miniconda3/lib/python3.8/site-packages/xgboost/lib/libxgboost.so(+0x1aa93d) [0x7fd82e1f793d]
[bt] (3) /miniconda3/lib/python3.8/site-packages/xgboost/lib/libxgboost.so(+0x1aac2d) [0x7fd82e1f7c2d]
[bt] (4) /miniconda3/lib/python3.8/site-packages/xgboost/lib/libxgboost.so(XGBoosterPredictFromDMatrix+0x346) [0x7fd82e0
```

```
import numpy as np

predictor.content_type = "text/csv"

# ☒ Model expects 4 features (columns)
x_test = np.random.randn(1, 4) # one row with 4 columns

# Convert to CSV string with newline at end
payload_csv = ",".join(map(str, x_test.flatten().tolist())) + "\n"

print("Payload being sent:\n", payload_csv)

response = predictor.predict(payload_csv)
print("☒ SageMaker Response:", response)
```

```
Payload being sent:
-0.07963663356745397,0.4164817166343696,0.2029543248682189,1.3311177155269787
```

```
☒ SageMaker Response: b'0.2985784113407135\n'
```

```
import time
sagemaker_latencies = []

for _ in range(20):
    start = time.time()
    predictor.predict(payload_csv)
    end = time.time()
    sagemaker_latencies.append((end - start) * 1000)

print(f"☐ SageMaker Inference Latency: {np.mean(sagemaker_latencies):.2f} ms")
```

```
☐ SageMaker Inference Latency: 197.03 ms
```

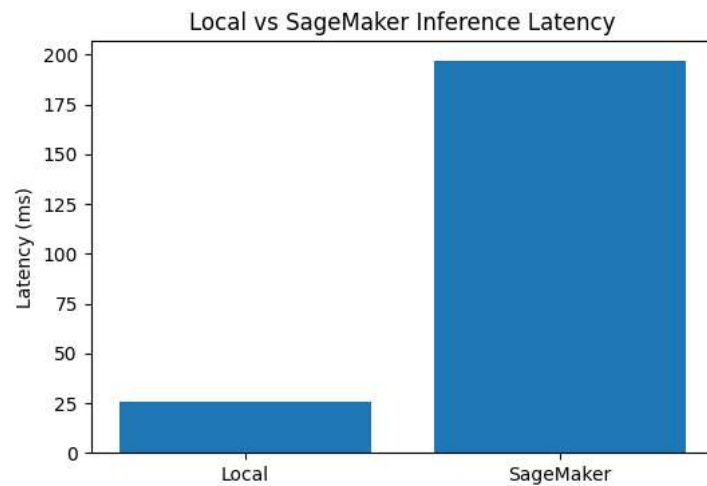
```
import matplotlib.pyplot as plt
import numpy as np

# Replace with your actual local latency from earlier
local_avg_latency = 25.4 # ms
sagemaker_avg_latency = 197.0 # ms

labels = ['Local', 'SageMaker']
latencies = [local_avg_latency, sagemaker_avg_latency]
```

```
plt.figure(figsize=(6,4))
plt.bar(labels, latencies)
plt.ylabel('Latency (ms)')
plt.title('Local vs SageMaker Inference Latency')
plt.show()

print(f"📊 Comparison:")
print(f"Local Inference Latency : {local_avg_latency:.2f} ms")
print(f"SageMaker Inference Latency: {sagemaker_avg_latency:.2f} ms")
print(f"Overhead due to Network/Container: {sagemaker_avg_latency - local_avg_latency:.2f} ms")
```



📊 Comparison:
Local Inference Latency : 25.40 ms
SageMaker Inference Latency: 197.00 ms
Overhead due to Network/Container: 171.60 ms

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